NOCR: End-to-End Structured Data Extraction from Historical Documents Using VLMs

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Abstract

Extracting structured data such as tables from historical archives is a time-consuming bottleneck in economic history research. Traditional approaches require optical character recognition (OCR) followed by manual postprocessing to structure the output. This two-step process is both inefficient and error-prone. This paper shows that vision language models (VLMs) can collapse this pipeline into a single step, directly converting scanned images of historical documents into structured JSON output. This end-to-end approach is both highly accurate and drastically reduces the need for post-processing. Using Google's Gemini and Gemma models on handwritten Dutch tax records from 1899, we evaluate performance across multiple model versions (2.0, 2.5, 3.0, 3.1, and open weight models) and prompting strategies (zero-shot and few-shot). Our results suggest that this VLM-based extraction, which we call NOCR, is already the superior choice for large-scale historical data projects, with clear evidence of rapid model improvement over time.[1]

[1]We thank William Skoglund, Jakob Molinder, and Ruben Peeters for their valuable input. All code to run and explore the experiments is available at https://github.com/HIP-NL/nocr-experiments.

1. Introduction

Economic history is centered on the quantitative analysis of archival records, from tax assessments and census data to account books, probate inventories, and notarial records. Often these sources are structured, that is, they are laid out as tables, lists or similar forms. The digitization of these sources has for a long time been a manual process, but is increasingly being automated. A common pipeline involves scanning physical documents, then processing them through optical character recognition (OCR) or handwritten text recognition (HTR) systems, and, finally, postprocessing to make the raw text output suitable for analysis.

This two-step approach has significant limitations. While traditional OCR/HTR systems already excel at transcribing even handwritten text for some time (Sang 2024; Leifert et al. 2024), they produce unstructured output that requires extensive manual postprocessing. When working with tabular or structured data, this step can consume more time than the initial transcription. Researchers must write custom parsers (e.g. Cummins 2021), handle implicit information (such as repeated values in table columns), and correct systematic errors in the OCR output.

The main alternative described in the literature is another two-step approach, where first a segmentation model is trained to divide the scans in sections that map to the fields that the researchers want to extract. Next, standard OCR tools are applied to these segments. This approach has been used, for example, by Dahl et al. (2023) to transcribe nurse journals or by Blomqvist et al. (2022) to digitise a Swedish wealth tax of 1571 Bailey et al. (2023). A toolkit for this approach is described in Shen et al. (2021). While most authors report good performance, setting up and maintaining a two-step approach is potentially more complex compared to a single model approach as suggested here.

Recent advances in vision language models (VLMs) offer an alternative by directly converting scanned images into structured data. Rather than extracting text and then inferring structure, VLMs can be prompted to understand both the visual layout and semantic content of a document simultaneously. We call this approach "NOCR", because we see it as a significant departure from traditional OCR methods. This approach even allows VLMs to handle complex cases that traditional OCR pipelines struggle with. Examples of this include splitting cell content in multiple fields, dealing with cells that implicitly copy values from rows above, or handling annotations written in the margins. VLMs furthermore have the capability to handle printed, typewritten, and handwritten text in many languages.

A further advantage is simplicity. A single API call replaces a multi-stage pipeline. Instead of maintaining separate OCR and postprocessing code, researchers can describe their desired output format in natural language and receive structured JSON in response, which can easily be converted to tabular data for downstream statistical analysis. This approach is particularly well-suited to the kind of historical documents analysed in economic history, where the visual context of the information often matters.

In addition, VLMs also support in-context learning, where inputs and the expected output are given to the models of examples. This approach can significant improve the accuracy of the output on the final prediction (Agarwal et al. 2024). We show below that this is a cost-effective strategy when using commercial model providers, as input tokens are usually priced much lower than output tokens. In this way, we have been able to use this approach to extract the tax data from archival scans in the HIP-NL project (Rijpma et al. 2025).[2], handling 10,000s records at very low cost, including typewritten, printed and handwritten records.

Related work includes Kim et al. (2025), who test a number of VLMs on early twentieth century probates as an alternative to traditional OCR. While they also find superior performance of VLMs in terms of lower CERs, they do not assess VLM's abilities to extract structured data directly, rather focusing on line-by-line transcription. Griesshaber and Streb (2025) also use VLMs to create datasets from late nineteenth and early twentieth-century German patent records. While they use VLMs throughout their pipeline, the text and feature extraction are separate steps. Moreover, we focus on the harder case of handwritten records, compared to the printed text used in the patents. In addition, we strongly emphasise the value of in-context learning as a strategy to increase accuracy while keeping cost under control, a point only made in passing by Kim et al. (2025). Ribeiro et al. (2026) finetune an early local VLM, DONUT, which we see as an important alternative for larger-scale cases. However, finetuning will not be a cost-effective strategy for all types of data. Probably, many researchers working with scanned text sources have already discovered similar approaches independently or through the grapevine.

In this study we demonstrate the viability of VLM-based extraction using Google's Gemini Flash family of models (Gemini Team et al. 2023) applied to handwritten Dutch tax records from 1899. Our evaluation addresses four questions. First, how well do VLMs perform at combined OCR and structuring tasks on historical documents (accuracy)? Second, what are the cost-performance tradeoffs across different models and prompting strategies? Third, are these models improving, and what does this suggest for future applications? Fourth, are open weight models a viable approach?

Six commercial model variants are tested from Google's Gemini range of models, across two prompting strategies (zero-shot and few-shot with examples) and extended reasoning modes. In addition, lighter comparisons are made with various sizes of open weight variants. Experiments are done on a dataset containing 40 individual tax records with ground truth annotations. In addition, the favoured model is applied to the full 1899 tax records, containing over 10,000 households. The results suggest that VLM-based extraction is not only viable but often superior to traditional approaches, particularly when total workflow time is considered.

[2] https://www.hipnl.nl/

2. Data and Methods

2.1 The Utrecht 1899 Tax Records

Our test dataset comes from the municipal archives of Utrecht in the Netherlands (Het Utrechts Archief).[3] The records document household income tax assessments for the year 1899, with each entry recording the household head's initials and surname, address, tax bracket, and taxes due. The appendix provides an example scan and record.

While these records are fairly clear and consistent as far as historical archival material goes, the documents present several challenges typical of historical economic data. First, they are handwritten and have corrections in the cell in a second hand. Names and addresses use 19th-century Dutch spelling conventions and abbreviations. Titles like "Wed." (widow), "Mej." (miss), and "geb." (born, indicating maiden name) must be correctly interpreted and extracted in a standardised way. Moreover, multiple variables are combined in a single cell, such as title, initials, surname, and maiden name, which we want to split. Finally, the numeric tax columns use different conventions from the other columns. For streets, empty cells indicate repetition, while an empty final column means "00" cents.

Our ground truth consists of four scanned pages (two double-page spreads, split into left and right images) containing 40 individual records, with data split over five columns. Each record includes nine fields that we want to extract: serial number (volgnummer), title, initials, surname, maiden name, street, house number, tax class, and tax amount. Note that these are to be extracted directly, with little to no postprocessing. Ground truth transcriptions were created by manual review. While the total number of scans is low, there are enough cells and characters to calculate meaningful performance metrics, while keeping the cost of running multiple experiments to a minimum.

2.2 Experimental Design

model name	short description
gemini-2.0-flash-lite	Lightweight, lowest cost
gemini-2.0-flash	Standard 2.0 version
gemini-2.5-flash-lite	Lightweight 2.5 generation
gemini-2.5-flash	Standard 2.5 version
gemini-3-flash	Standard 3.0 generation
gemini-3.1-flash-lite-preview	Lightweight 3.1 generation

We evaluated six Gemini Flash models. For each model, we tested multiple configurations. For prompting, we tested a zero-shot (prompt and target image only) versus few-shot (three example images with ground truth JSON, followed by target

[3] https://hetutrechtsarchief.nl/collectie/3155488E231AF282E0638F04000A85F3

image) strategy. Few-shot or in-context learning allows the model to learn formatting conventions and error patterns from examples, at the cost of longer context windows and higher input token usage.[4]

We also tested for the impact of standard inference versus extended thinking (2000-token budget). The thinking mode allows models to generate internal reasoning chains before producing output, potentially improving accuracy on complex tasks. This comes at the cost of higher output token usage.

Our prompt provides detailed instructions for each field, including specific guidance on edge cases (e.g., "If a street is omitted because it is the same as the previous row fill in the missing value with the last explicitly stated one"). The prompt was iteratively built to try and catch obvious errors in the output. The full prompt is provided in Appendix B.

We use a leave-one-out evaluation design: for few-shot experiments with four total images, we use three as examples and the fourth as the target, rotating through all combinations. This ensures each image is evaluated exactly once per model-strategy combination. The Gemini API is stateless, so there is no contamination of the evaluation data.

2.3 Evaluation Metrics

We evaluate our approaches using three metrics. As a fine-grained measure of transcription accuracy, we use the *Character Error Rate (CER)*. For each record, we calculate the Optimal String Alignment (OSA) distance between predicted and ground truth values across all nine fields. For this we use the `stringsim` function from the R `stringdist` package (Loo 2014). The OSA distances counts the total number of deletions, insertions, substitutions, and transpositions needed to change the original model response in the ground truth data.

Because our goal is to create tabular output, we also calculate the *Cell Error Rate* as the share of fields (out of nine per record) that contain any error. Here a field with a single incorrect character counts the same as a completely wrong field. This metric better reflects the downstream cost of errors, since even small mistakes can require manual correction. It should however be noted that some cell errors are not as consequential as others. For example, a one-character mistake in a surname is arguably not that important if the names are not used for the analysis, or if they are processed in an ML-based record linkage approach that allows for small spelling variations (e.g. Rijpma et al. 2020). We do not use the tree-edit distance-based similarity (TEDS, proposed in Zhong et al. 2020) as its key strength, handling cell-alignment issues in tables, is not needed here as cells are never misaligned in our experiments' output.

We also calculate the total *Cost* per page based on Gemini API pricing (as of January 2025), accounting for input tokens, output tokens, and thinking tokens where applicable. Prices range from $0.075 per million input tokens (flash-lite) to $3.00

[4]Niels Rogge for the idea to do this: https://x.com/NielsRogge/status/1901373186043998525.

per million output tokens (flash-3).[5]

All 40 records are evaluated for each model-strategy combination, giving 800 total scan-level predictions and 7200 predicted cells across our experimental design.

3. Model Performance

3.1 Overall Accuracy

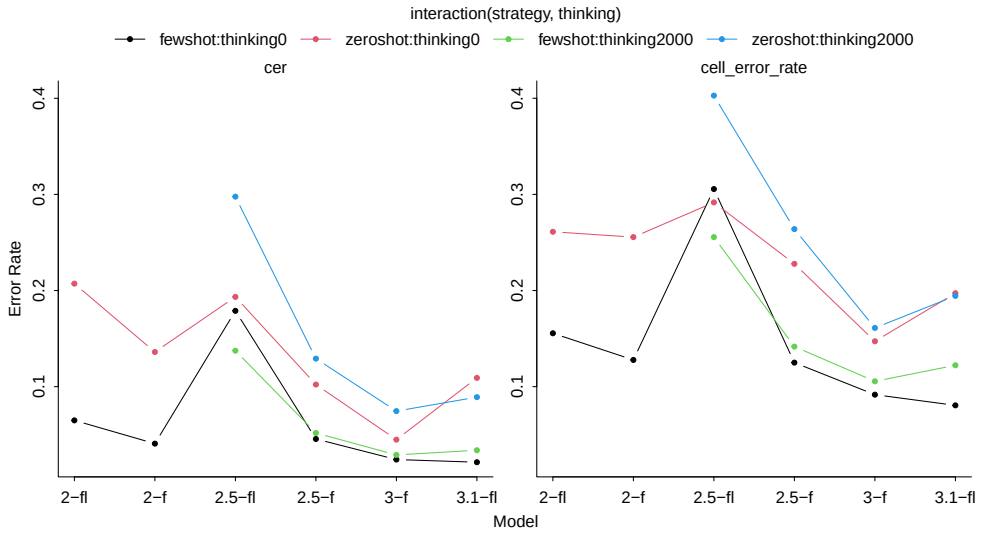


Figure 1: Error rates by model and strategy

Figure 1 shows character and cell error rates across all model-strategy combinations. The first thing to observe is that the trend in model performance improves over the generations. Even if some models deteriorate over previous generations in some strategies, overall we should expect newer models to perform better.

Few-shot learning is shown to provide substantial benefits. Across all models, few-shot prompting strongly improves error rates. For 3.0-flash, few-shot reduces CER from 4.5% (zero-shot) to 2.3% (few-shot). The effect is even stronger for earlier model versions—Gemini 2.0's CER drops from 13.2% to 4.0% with few-shot examples. The best error rate, 2.1% is obtained using few-shot prompting for the 3.1-flash-lite model. This suggests that showing the model correct formatting is very valuable. Interestingly, zero-shot Gemini 3.0 outperforms few-shot Gemini 2.0 and 2.5, suggesting that model quality can substitute for in-context learning to some degree. However, the combination of best model and few-shot prompting remains superior overall.

Extended thinking shows poor results. The 2000-token thinking budget produces inconsistent effects. For all models except 2.5-Flash-lite, thinking slightly degrades

[5]https://ai.google.dev/gemini-api/docs/pricing. Unfortunately, the generous free tier of the Gemini API (200 requests, one million tokens per day) is no longer available.

Table 2: Top 5 models and strategies by error rate.

model	strategy	thinking	cer	cell_error_rate
gemini-3.1-flash-lite-preview	fewshot	thinking0	0.021	0.081
gemini-3-flash-preview	fewshot	thinking0	0.024	0.092
gemini-3-flash-preview	fewshot	thinking2000	0.029	0.106
gemini-3.1-flash-lite-preview	fewshot	thinking2000	0.034	0.122
gemini-2.0-flash	fewshot	thinking0	0.041	0.128

performance. Notably, 3.0-Flash often refuses to use extended thinking even when prompted, reverting to standard inference. This suggests that VLMs do not need explicit chain-of-thought to extract data from relatively straightforward and consistent visual inputs. As thinking modes consume more expensive output tokens, these results suggest that the use of thinking by the models should be minimised.

Lite models show varying performance, with standard models often outperforming their "lite" counterparts. This suggests that model compression strategies such as distillation and quantisation can hurt performance (see also section 6 comparing quantised with non-quantised models). As the lite models are also the fastest and the cheapest, this has important implications for large-scale and cost-sensitive applications. At the same time, it should be noted that the 3.1-Flash-Lite model is the best performing one in combination with a few-shot prompting strategy.

The top five configurations ordered by character error rate are listed in table 2. Looking at the levels, even the best-performing model still makes errors on roughly one in twelve cells. The best performance model gets roughly one in every 50 character wrong. These numbers are in line with, and often improve on traditional OCR methods for historical handwriting [6] While error rates are still high enough that output requires review, in our experience the NOCR process is vastly more efficient than manual transcription from scratch.

Examining specific errors reveals predictable patterns. Common mistakes include confusing '0' and '8' in sequential numbers and house numbers (e.g., predicting 1061 vs ground truth 1861), incorrectly handling of implicit street name repetition, and errors in parsing decimals separated by a vertical bar in tax amounts.

That these errors seem to cluster around specific challenging cases rather than being randomly distributed suggests that improvements are possible through more detailed prompt engineering, fine-tuning on similar documents, or automated error catching. We further assess error types in the full 1899 data, when we have enough observations to systematically catch the most important patterns.

[6] E.g. 3.7% on typewritten text by Amujala et al. (2022); 8.4% on 16th century handwriting by Blomqvist et al. (2022); 5.5% on nineteenth-century handwriting by Sang (2024); 5% by Muehlberger et al. (2019); 3% by Koert et al. (2024).

4. Cost-Performance Analysis

4.1 The Cost-Accuracy Frontier

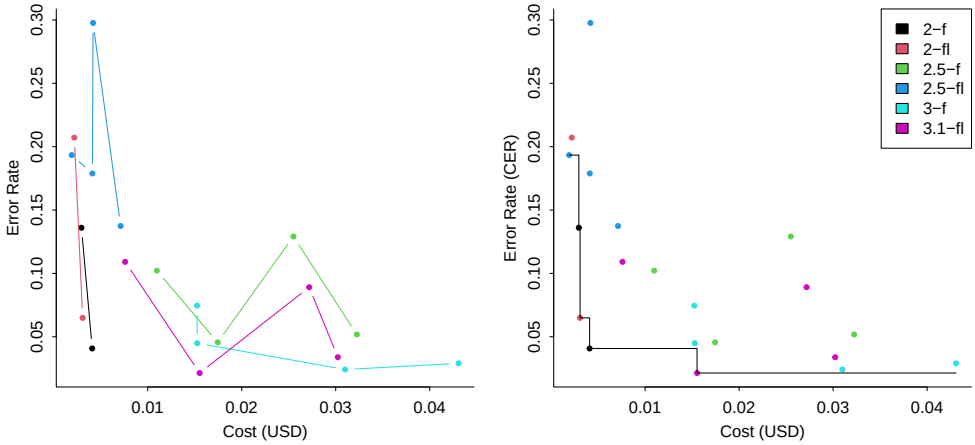


Figure 2: Error rates by cost and model. In the left figure, the connected points for each model indicate increasingly expensive strategies: zero-shot/no thinking, zero-shot/thinking, and few-shot/thinking. The right figure shows the same data but with the Pareto cost-performance frontier visualised.

Figure 2 plots error rates against total cost per page for all configurations and table 3 lists the best-performing models by cost. Cost varies dramatically across configurations, from approximately $0.002 per page (Gemini 2.0 Flash Lite zero-shot) to $0.043 per page (3.0-Flash few-shot with thinking). This 20-fold range can accommodate different use cases. The cheapest models combined with a few-shot strategy (2.0 Flash or 2.0 Flash Lite) can achieve strong performance (4.4 and 6.5% CER) for low prices ($0.004 and $0.003 per page respectively). Better performance can be obtained with a higher budget. The 3.0-Flash models provide high performance, but at the highest prices (c. 2.5% at $0.03–0.04 per scan). We think the optimal model in these experiments is the combination of a 3.1-flash-lite model with few-shot prompting, providing the highest performance at a mid-range budget (0.016). The cheapest models here could process a thousand scans pages for about $3; the mid-range option would cost $16, a fraction of manual transcription costs, with accuracy already sufficient for most quantitative analyses. To put those pages counts into perspective, the full 1899 tax records for Utrecht adds up to 1108 scans; by 1920, that number had grown to 2828.

4.2 Scaling Considerations

Several factors could further reduce costs at scale. First, context caching could reduce costs. If the few-shot examples remain constant across pages from the same document type. Gemini's context caching feature could reduce input token costs by up to 75% for few-shot scenarios, making the more expensive models more af-

Table 3: Model cost and performance for models with with at least ten percent CER, ordered by costs

model	strategy	thinking	cost	error_rate
gemini-2.0-flash-lite	fewshot	thinking0	0.003	0.065
gemini-2.0-flash	fewshot	thinking0	0.004	0.041
gemini-3-flash-preview	zeroshot	thinking2000	0.015	0.075
gemini-3-flash-preview	zeroshot	thinking0	0.015	0.045
gemini-3.1-flash-lite-preview	fewshot	thinking0	0.016	0.021
gemini-2.5-flash	fewshot	thinking0	0.017	0.046
gemini-3.1-flash-lite-preview	zeroshot	thinking2000	0.027	0.089
gemini-3.1-flash-lite-preview	fewshot	thinking2000	0.030	0.034
gemini-3-flash-preview	fewshot	thinking0	0.031	0.024
gemini-2.5-flash	fewshot	thinking2000	0.032	0.052
gemini-3-flash-preview	fewshot	thinking2000	0.043	0.029

fordable.[7]

Likewise, processing documents in batches could enable further optimizations and volume discounts. This reduces costs by 50%. It is straightforward to set up, and the only downside is a delay of up to 24 hours. We have not used this feature or caching in these experiments.

Mixed strategies could also be valuable. Projects could use expensive models for challenging documents and cheaper models for routine cases, optimizing the overall cost-accuracy tradeoff. This would require a reliable method to assess page difficulty. If the routine model is cheap enough, low quality output might be detected after inference and sent back to more powerful models.

VLM pricing trends are also an important factor. There have been substantial reductions, e.g. the costs of 2.5 at its introduction was a factor ten lower than 2.0 at its introduction. Moverover, the overall trend in LLM pricing is negative, decreasing by c. 25% each year. At the same time, if we omit the introduction pricing of 2.5 compared to 2.0, each Gemini Flash model was more expensive than the one that came before.[8] Moreover, the generous free tier of the Gemini API has been removed.

A final way to economise on costs is to experiment with the output format. JSON, as used here, is heavy on brackets, quotation marks, and tabs, which all cost tokens. In addition, the key is repeated for every observation. By contrast, YAML and markdown tables have far better schema efficiency and have lower punctuation overhead, and these formats should also feature heavily in the models' training distribution. To explore this, we experimented with different output formats for two high-performing models (3.1-flash-lite and 2.5 flash) at different thinking budgets. The results show that markdown and YAML use 48% and 73% of the to-

[7] https://ai.google.dev/gemini-api/docs/caching?lang=python.

[8] https://github.com/sanando/llmpricing. No scholarly review of LLM pricing trends could be found at the time of writing.

kens that JSON does in our examples. Overall costs are not impacted to the same degree in our setting, as the (few-shot) prompt and thinking budgets make up 50–90% of the budget. Performace is also affected, with YAML outperforming JSON, and markdown doing worse (see appendix). In addition, it should be noted that the markdown output required cleaning up explanatory text before and after the table in 11 out of 12 results; and that over half the YAML results omitted underscores in the keys that needed to be fixed. A notable advantage for JSON, though token-inefficient, is that it is directly supported as an output format, and that it even possible to enforce schema adherence using libraries like pydantic.[9]

5 Performance on the full 1899 Utrecht tax data

To assess the viability of the NOCR workflow at scale, we report on our application of the pipeline to a complete corpus of scans from the 1899 Utrecht tax records, the full version of the data we used for the experiments above. In addition to showing the viability, it can also highlight some practical lessons. This dataset consists of 10,659 households spread over 1074 scans. This task was done in April 2025, using Gemini 2.0 Flash with few-shot prompting. The outputs generated by the VLM were subsequently checked and corrected by research assistants before adding them to the HIPNL database.[10] The resulting manually verified tables can serve as ground truth data, allowing us to evaluate the performance of our few-shot prompting strategy on the case of producing a large research-grade database.

To quantify the performance of our approach, we evaluated the data on a by-column basis using string similarity scores. Here we use the Levenshtein distance (so omitting transpositions compared to the experiments above) as access to the specific transformations in this implementation will allow us detect error patterns below.

For most fields, such as surnames, titles, tax classes, and tax amounts, the mean similarity scores between the VLM's raw output and the RAs' corrected ground truth are around 0.95 or higher, and cell error rates are generally low, indicating highly accurate transcription (table 4). Performance on the initials field is somewhat lower, indicating that the model, not unlike an inexperienced human transcriber, struggles somewhat with historical handwritten capital letters without the context of the rest of the word.

Overall, the performance of the approach in a project like this is highly encouraging. We achieved a mean Character Error Rate (CER) of 12.22%, alongside an overall cell error rate of 17%. As will be discussed below, the higher CER compared to the model's performance above is due to the handling of the street colum. Omitting this column results in a CER of 4.2%, in line with the results of the experiments above.

[9] https://github.com/pydantic/pydantic. In our experience JSON schema adherence is good enough that this step can be omitted in practice.

[10] In addition, the NOCR approach was also used for all printed and typewritten records. As these are far easier for the model, performance was superior to the results presented here.

Table 4: Mean Similarity Scores and Cell Error Rates per Column

Column	Mean_Similarity	Cell_Error_Rate
class	0.954	0.053
house_number	0.927	0.111
initials	0.928	0.175
street	0.788	0.465
surname	0.944	0.211
tax	0.948	0.122
title	0.982	0.021
volgnummer	0.929	0.202

Table 5: The 10 most frequent mistakes in the automated transcription of the 1899 Utrecht tax records

Field	Original	Corrected	Count
Tax	0.25	8.25	335
Tax	10	18	285
Class	1	7	69
Initials	A.	M.	58
Initials	A.	H.	46
Class	21	2	41
Class	1	3	36
Class	1	2	36
Tax	80.5	88.5	32
Tax	100	180	31

Perhaps the most important consideration is that the raw JSON responses returned by the model could be converted into structured, tabular data "as is". We did not implement any post-processing pipelines to handle formatting. RAs could go straight to work in the tables and potential mistakes could be flagged beforehand (e.g. and discrepancies between the tax due and tax bracket columns). The required manual corrections were generally minimal, the most important one probably being correcting 0-8 mistakes (table 5).

This poor performance in the street column is largely due to the structural logic of the documents causing significant cascading errors. In the original 1899 ledgers, clerks left the street column empty for consecutive households living on the same street. To create a complete dataset, we instructed the model to carry these street observations forward when encountering empty spaces. As a result, mistakes in this column like pushing a maiden name into the address column or even an technically correct one such as filling in a new address for one specific household that was added in pencil would be pushed down into multiple empty cells (see appendix). In cases like this, a single extraction error punishes model performance across multiple rows, artificially inflating the error rate for this column relative to the actual

transcription capability of the model.

To further investigate error patterns, we used the "trafos"option in the `adist()` function in R to break down the string distance in the individual operations needed to optimally change one string in another. We use this to find series of at least four deletions and insertions. This allows us to separate structural mistakes like omitting words from the correct column or inserting words from another field from character-level mistakes. We identify these structural mistakes by searching for long, consecutive blocks of deletions or insertions in the transformation strings.

Table 6: Distribution of Error Types (Structural Transformations) per Column

Column	None	Transcription	Omission	Inclusion	Omission and inclusion
class	10089	565	0	0	0
house_number	9476	1119	30	29	0
initials	8791	1840	6	17	0
street	5698	4258	349	344	5
surname	8404	2072	137	41	0
tax	9352	1301	0	1	0
title	10427	189	31	7	0
volgnummer	8504	2150	0	0	0

Table 6 shows that such structural mistakes mostly occur in the street column. The fill-down prompt dynamic creates nearly 700 combined structural errors in `street`, compared to much lower frequencies in other textual or numerical columns. For numeric columns like `tax` and `class`, structural mistakes are practically nonexistent, remaining limited to small transcription errors.

6. Open models

To keep transcription costs down and ensure that research stays reproducible, it would be desirable that open weight models can perform data extraction tasks. To investigate this, we look at the Gemma 4 class of models. These models share architecture, data, and training strategies with the Gemini models tested above (Gemma Team et al. 2024).[11].

The experimental setup is largely the same as above: we test performance using the CER, comparing zero shot and few shot prompting strategies across a range of models. Formats and thinking budgets are not tested. Because we know the number of parameters for these open weight models, we can directly assess how performance changes as model size varies. The largest two models were run on the Gemini API. Given the cash-strapped nature of my university, I only had access to a GTX2080 GPU, meaning that I could only run the smallest 2B models locally in unquantised form. For quantisation, we relied on the Quantization-Aware Training (QAT) models in 4-bit integer form. The quantised models are reported separately, and

[11]https://huggingface.co/collections/google/gemma-4

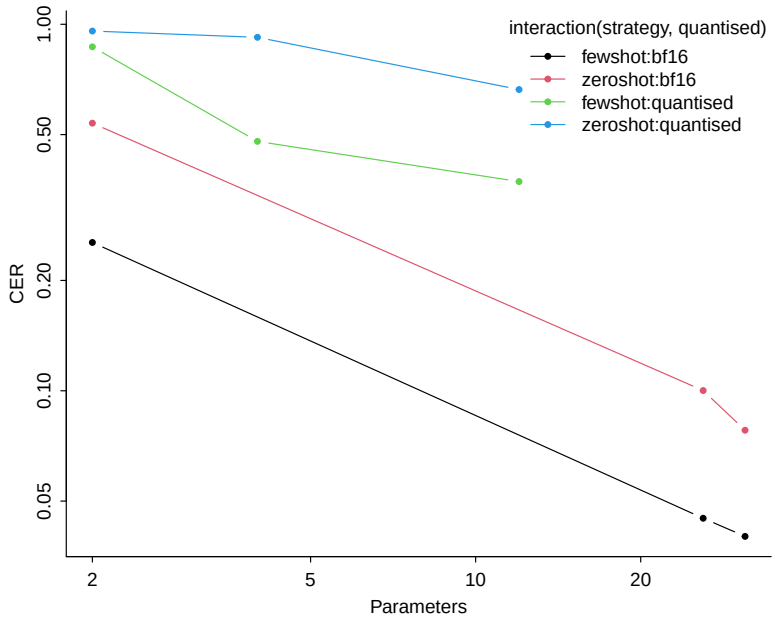


Figure 3: Performance and parameter size of open weight Gemma models, by prompting strategy and model quantisation

the impact of this is by itself of interest as locally run models will often have to be quantised.

The good news is that best open models tested are in fact competitive with commercial, closed models. Gemma 4 32B 26BMoE, using a few-shot setup would rank 5th and 7th on performance in terms of CER, and the accuracy is similar to the models we used for the full 1899 tax data (4% from Gemini Flash 2.0 in a few shot approach).

However, these are heavy models that cannot easily be run locally. Ingesting the images, in particular in the few-shot approach, blow up the VRAM requirements. Looking at performance across model sizes, we can see that performances deteriorates linearly as log parameter count decreases. Few-shot prompting again improves performance across model sizes and quantisation. It can also be seen that the quantised models perform much worse, and don't improve as strongly as parameter size increases as the non-quantised models. In these models we frequently saw hallucinated observations which greatly increased the CER.[12].

In short, when it comes to the direct extraction of handwritten structured data, sufficiently large open-weight models can substitute for closed commercial models. However, the small quantised models that can easily be run locally do not seem up to the task. These would probably need extensive finetuning to work.

[12]Older, small and quantised models like Qwen3-VL-4B-Instruct-3bit could not even process the few-shot approach correctly, either repeating the examples or output complete gibberish

7. Discussion

This paper demonstrates that VLM-based extraction of structured historical data is not just viable but in fact competitive with traditional approaches, and can even outperform it. The most important finding is that direct scan-to-JSON works reliably. Even complex tabular layouts with implicit data and historical handwriting can be accurately *and directly* parsed into structured output without separate OCR and postprocessing steps. Applying this to large corpuses of sources results in databases that are easily parsed and checked. Error rates overall are workable, though care should be taken in dealing with the logical structure of the scans. In our Utrecht 1899 case, most of the error rate could be traced to the decision to the let the model take care of the last-observation-carried-forward logic in the scans, which resulted in carrying forward mistakes as well.

A further conclusion is that few-shot learning is a highly effective way to improve performance. The dramatic improvement from showing just three examples suggests that in-context learning remains valuable even as base model capabilities improve. Moreover, this approach makes relatively cheaper models interesting options.

Cost-performance tradeoffs are also favorable. For $0.004 per page, we achieve error rates low enough to support quantitative analysis with modest manual review. This is orders of magnitude cheaper than manual transcription (estimated here at $1–5 per page for professional transcribers). Few-shot prompting also provides some of the most interesting price-performance points as the input tokens for the extented prompt are usually priced far lower than output tokens.

Our results show that performance is improving rapidly. The 65% reduction in error rates from Gemini-Flash 2.0 to Gemini-Flash 3.0 (released less than a year apart) suggests a steep improvement curve. Gemini-3.1-flash-lite-preview, released in March 2026, presented another improvement compared to 3.0. The flipside to this is that models are deprecated rapidly as well. At the time of writing, the Gemini 1.5 models that were originally part of our tests and setup for printed documents are no longer available through the API.

There are caveats to our findings. We evaluate only one document type (Utrecht tax records from one year). Generalization to other formats, languages, or time periods requires further testing. We would suggest large projects go through similar experiments as outlined here to find the optimal model and strategy for their use case. Anecdotally, we have found this approach to work well on most printed, typewritten, and handwritten records from the eighteenth century onwards. A key constraint is that we have found that applying these methods to pages with tables consisting of many rows without gridlines can result in output where the columns are out of sync which can take a lot of time to fix.

Open source models are a promising avenue for further research. Large open models with an architecture similar to the commercial models tested here were shown to be viable alternatives. Moreover, thre is a very large number of open

weight VLMs are available that can be run locally (e.g. Qwen2-VL, Wang et al. 2024; PaliGemma Beyer et al. 2024). These hold the promise of eliminating API costs, pricing changes, data privacy concerns, vendor lock-in, model deprecation, all of which can disrupt projects. Fine-tuning smaller models on large corpora of historical documents (perhaps using cheaper models to generate noisy labels for expensive models to refine) could substantially improve accuracy on domain-specific challenges like historical handwriting and orthography. Our own experiments in finetuning the DONUT family of models were promising in this regard (Kim et al. 2022; see a follow-up in Ribeiro et al. 2026), but using commercial models with in-context learning through an API was more efficient for us. Doing a full fine-tune could require 100s of labelled scans for each municipality's particular type of tax records. Most municipalities did not have enough record to make that a cost-efficient approach compared to the setup described above.

Getting models to provide confidence scores for each extracted field would also be valuable. This would enable a hybrid workflows where high-confidence extractions are accepted automatically while uncertain cases receive manual review, or follow-up with a more powerful and expensive model. A simple but cost-inefficient way to do this would be to have multiple model runs and have them vote on the output.

Finally, the tabular tax records used represent one type of document in the vast space of historical records available in the archives. Applying similar methods to census records, notarial archives, merchant ledgers, and correspondence would demonstrate broader applicability.

The viability of VLM-based extraction has significant implications for historical research. As costs fall and accuracy improves, creating structured datasets from archival sources becomes feasible for individual researchers with limited funding. For better funded researchers, projects involving millions of pages become tractable. The key constrain becomes whether archives can provide their records as high-quality scans.

Appendix A: Sample Scans and Ground Truth

The scans containing the tax records look as follows:

Below is a sample ground truth record demonstrating the structure and complexity of the data:

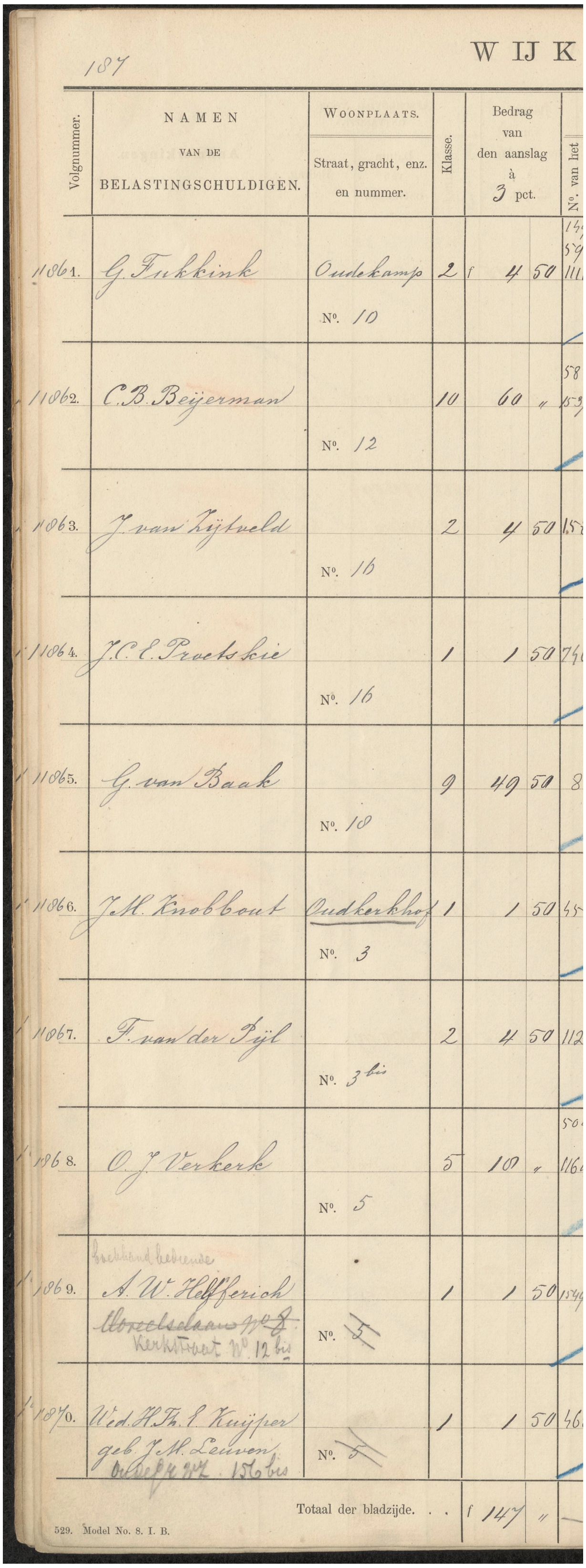

```
{
    "volgnummer": 1861,
    "title": null,
    "initials": "G.",
    "surname": "Fukkink",
    "maiden_name": null,
    "street": "Oudekamp",
    "house_number": "10",
    "class": 2,
    "tax": 4.5
}
```


This record shows a typical entry: sequential number 1861, no title, initials “G.”, surname “Fukkink” (note the challenging double-k and -ink ending), address at Oudekamp 10, tax class 2, and tax amount of 4.5 guilders (using decimal notation after conversion from the vertical-line separator in the original).

More complex cases include widows with maiden names:


```
{
    "volgnummer": 1870,
    "title": "Wed.",
    "initials": "H.Th.E.",
    "surname": "Kuijper",
    "maiden_name": "J.M. Leuven",
    "street": "Oudkerkhof",
    "house_number": "5",
    "class": 1,
    "tax": 1.5
}
```


Here “Wed.” (Weduwe, widow) as the title, multi-part initials “H.Th.E.”, and the maiden name “J.M. Leuven” (indicated by “geb.” in the original scan) demonstrate the richer structure that VLMs must correctly parse.



187 WIJK

Volgnummer.	NAMEN VAN DE BELASTINGSCHULDIGEN.	WOONPLAATS. Straat, gracht, enz. en nummer.	Klasse.	Bedrag van den aanslag à 3 pct.	N°. van het
11861.	G. Tukkink	Oudekamp N°. 10	2	4 50	111
11862.	C.B. Beijerman	N°. 12	10	60 "	153
11863.	J. van Zijtveld	N°. 16	2	4 50	15
11864.	J.C.E. Proetschie	N°. 16	1	1 50	74
11865.	G. van Baak	N°. 18	9	49 50	8
11866.	J.H. Knobbout	Oudkerkhof N°. 3	1	1 50	45
11867.	F. van der Pijl	N°. 3bis	2	4 50	112
11868.	O.J. Verkerk	N°. 5	5	10 "	116
11869.	A.W. Helfferich ~~Moreelselaan N°. 8~~ Kerkstraat N°. 12bis	N°. ~~5~~	1	1 50	154
11870.	Wed. H.Th.E. Kuijper geb. J.M. Leuven	N°. ~~5~~	1	1 50	46
	Totaal der bladzijde. . .			147 "	

529. Model No. 8. I. B.

Figure 4: Handwritten municipal income tax records from the Utrecht Archives, 1007-2 Gemeentebestuur van Utrecht 1813-1969, no 6613

Appendix B: Prompt

The full prompt used for all experiments:

```
Perform the following task using step-by-step reasoning.

Extract the data from this scan and return it as json.

The scan contains five columns describing individuals being taxed in
the Dutch city of Utrecht in 1899. Extract the following information
and return it as json:

- volgnummer, the sequential number of the individual in the scan,
  an integer number at most four digits long. If there are more digits,
  the first one is probably from the previous page and should be ignored.

- title, a title such as 'Wed.', 'Dr.', 'Mej', 'jkhr', 'jkvr', 'geb',
  'vert', 'p/a', 'cur'. If the scan does not contain a title, leave
  this empty (null).

- initials, the initials of the individual.

- surname, the surname of the individual.

- maiden name, the full maiden name of the individual, below the
  husband's name and indicated with 'geb' (for 'geboren'), typically
  provided if she is a widow.

- street, the street in Utrecht of the individual, always in the
  'woonplaats' column. If a street is omitted because it is the same
  as the previous row fill in the missing value with the last explicitly
  stated one. If the scan does not start with a street to fill in, leave
  the street empty. Below the name another address may be written if the
  individual moved. You should ignore this and only provide the original
  address from the woonplaats column.

- house_number, the house number of the individual, after the printed 'No.'.
  This is an integer number, sometimes followed by a letter (a, b), a
  superscript number, or a superscript text (usually 'bis'). Usually, the
  house numbers within a street will be increasing.

- class, the tax class of the individual, in the 'klasse' column', an
  integer number. This should always be transcribed. In the first row
  there will be an 'f' next to the class, which should be ignored.

- tax, the tax amount due for the individual, in the 'Bedrag van den
```

```
  aanslag' column. This is a decimal number, where a vertical printed
  line is used as the decimal separator. A quote mark can also follow
  the vertical line instead of the usual two digits, which should be
  transcribed as two zeroes. A little bit of the next column might be
  visible, but this should be ignored, only the part in the 'Bedrag van
  den aanslag' column should be transcribed.

Carefully distinguish between the digits '0' and '8'. The digit '0' is
a simple oval with a continuous outline, maybe open at the top if the
writer was in a hurry, while '8' consists of two stacked loops, typically
crossing the loop.
```

The prompt combines general instructions with specific edge case handling. Key features include explicit field descriptions, where each JSON field is described in detail with examples.[13] Where a field can only contain a number of options, these are specified. There is edge case guidance to instruct the model how to handle implicit data (street repetition), moved addresses, and formatting conventions. Character-specific instructions for the distinction between a ‘0’ vs ‘8’, though this was not very succesful. We explain column headers and layouts helps the model understand the visual organization. And finally, we request JSON to ensure structured, parsable output. The “step-by-step reasoning” prefix was included to encourage extended thinking, though as noted earlier, Gemini 3.0 often ignores this instruction when it considers reasoning unnecessary.

[13] In addition, it can also be useful to specify the datatype for each field.

Appendix C: Performance under different file formats

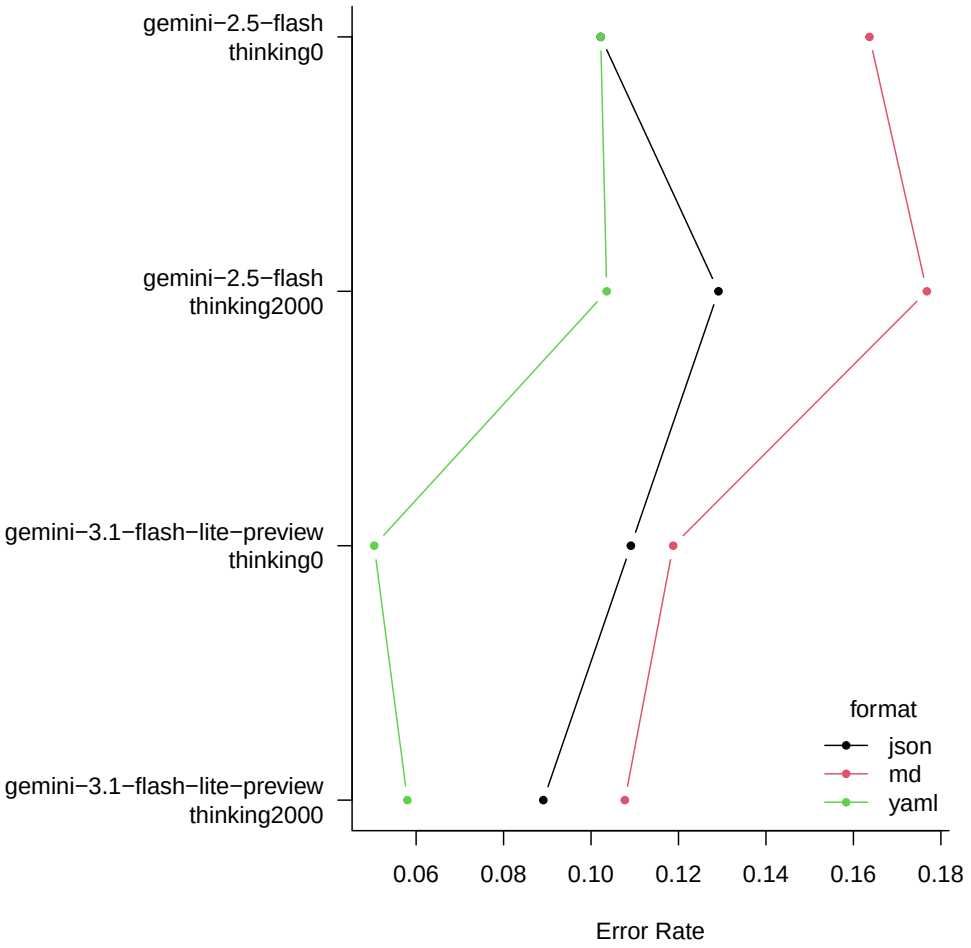
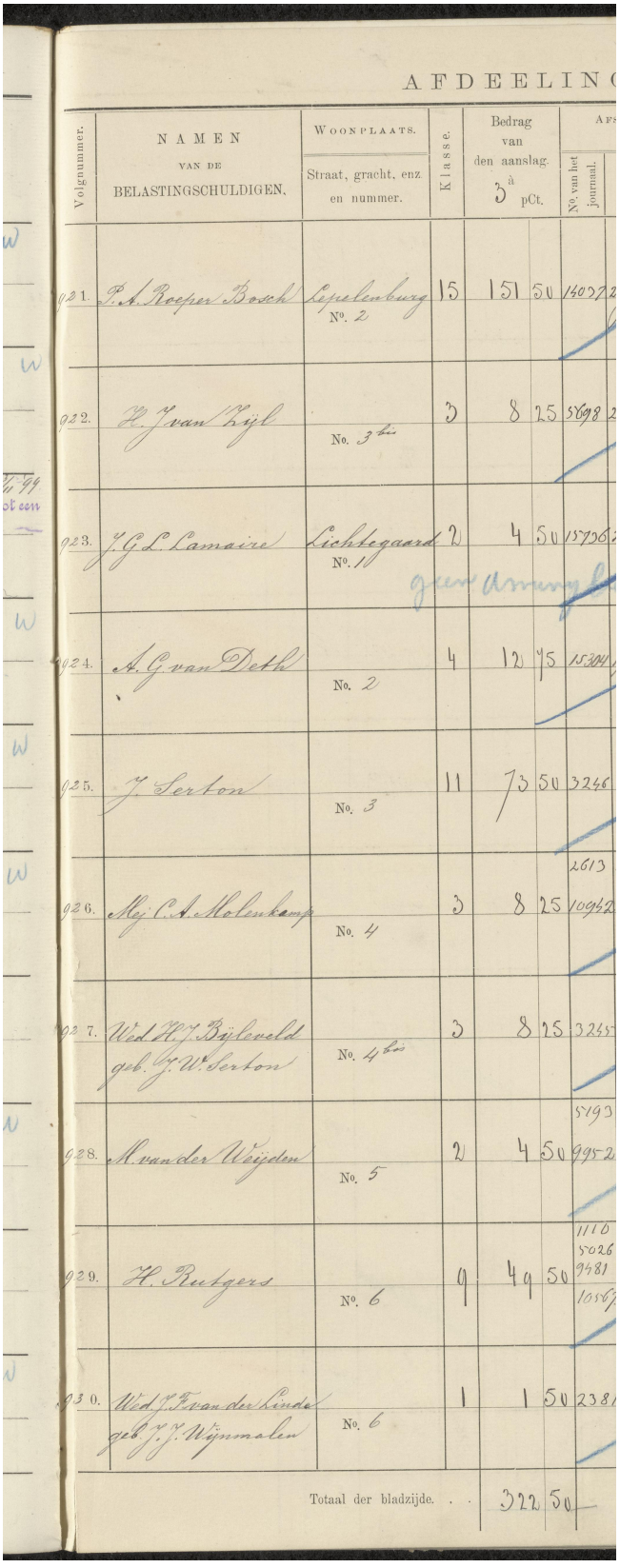


Figure 5: Character error rate (CER) by model, thinking budget, and output format. Note: strategy fixed to zero shot prompting.

Table 7: Example of carried forward, original json response and corrected data

street_orig	street_corr	street_sim
Lepelenburg	Lepelenburg	1.000
Lepelenburg	Lepelenburg	1.000
Lichthgaard	Lichtegaard	0.909
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273
Lepelenburg	Lichtegaard	0.273

Appendix D: Example of fill-down issues



AFDEELING

Volgnummer.	NAMEN VAN DE BELASTINGSCHULDIGEN.	WOONPLAATS. Straat, gracht, enz en nummer.	Klasse.	Bedrag van den aanslag. 3 à pCt.			Afs. No. van het journaal.
921.	P. A. Roepers Bosch	Lepelenburg No. 2	15	151	50		14027
922.	H. J. van Rijk	No. 3 bis	3	8	25		5698
923.	J. G. L. Lamaire	Lichtegaard No. 1	2	4	50		15736
924.	A. G. van Deth	No. 2	4	12	75		15304
925.	J. Serton	No. 3	11	73	50		3246
926.	Mej. C. A. Molenkamp	No. 4	3	8	25		2613 10945
927.	Wed. H. J. Bijleveld geb. Jr. W. Serton	No. 4 bis	3	8	25		3245
928.	M. van der Weijden	No. 5	2	4	50		5493 9952
929.	H. Rutgers	No. 6	9	49	50		1110 5026 9481 10567
930.	Wed. J. F. van der Linde geb. J. J. Wijnmalen	No. 6	1	1	50		2381
	Totaal der bladzijde. . .			322	50		

Figure 6: Example of fill down dynamic in Utrecht 1899 HO tax. Source: HUA 1007-2 no 6612

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